

Experiment 3: guide for the lab

10060 Physics (Polytechnical Foundation) · Friday 25 September 2026

Equipment from 13:00 · report due 17:00 · files: this PDF and Experiment3_analysis.ipynb

1 The short version

Part 1. Choose hypothesis C. Push the iOLab through its force probe while the accelerometer and the force probe record together. The force probe shows when the hand lets go: the force drops back to zero. Read the acceleration at that moment. It equals the flat friction level (point C). It is not the peak (A) and not zero (B). Newton's second law explains why: once the hand is gone, kinetic friction is the only horizontal force, so the acceleration is the constant $a = -\mu_k g$.

Part 2. Hypothesis: going up, the iOLab slows down faster than it speeds up going down, $|a_1| > |a_2|$, by more than three combined uncertainties. Measure a_1 and a_2 with the wheel sensor in 6 to 8 runs, average them, and compute a standard score. Theory: rolling resistance always opposes the motion. Going up it points down the ramp and adds to gravity. Going down it points up the ramp and subtracts from gravity.

What you hand in at 17:00

- One lab report of at most 3 pages, then page 4 with the reflections (3 to 4 lines for each of the three questions).
- A hand drawn sketch of each setup. Draw on paper, photograph it from straight above, crop it.
- Every measured value with its uncertainty and its unit.
- The DTU Learn upload folder is set up per group. Agree now who uploads.
- You write the text yourselves. The Experiment 1 sheet says you may not use AI to write the lab report; assume the same rule today. This guide and the notebook give you the physics, the method and the numbers. The sentences are yours.

What to bring

Laptop with the iOLab app and Python (numpy, pandas, matplotlib), this guide, the notebook, a ruler or tape measure, tape, paper and pen for the sketches and the run log, and a phone for photos. Charge the laptop.

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2 The rubric, line by line

Every row of the rubric has four levels. This table says what level 4 asks for and what you do to get it.

Rubric line	Level 4 wording	What you do
Sketch of the setup	A very good and clear sketch.	One hand drawn sketch per part. Labels: iOLab orientation, the y axis and its positive direction, the force probe, the push direction, the surface; for the ramp θ , H , L , the measuring wheel, up and down, where you catch it. Section 9.3 has the full list.
Purpose and communication	Clear purpose, good communication, all measurements with units.	Purpose in one or two sentences at the top of each part. Every number with uncertainty and unit. State the sensors, the sample rates and the number of runs.
Data	All data collected, organized and presented clearly.	One table per part with every run, not only the mean. One figure per part with axis labels, units, marked points and a caption.
Hypothesis	Formulated, clear and testable.	Name the measured quantity, the predicted relation, how you measure it, and the number that decides ($z \leq 2$ or $z > 3$).
Validity of the hypothesis	The conclusion refers to the data and results.	Write “rejected” or “not rejected”, quote the numbers with uncertainties and the z value, then give the physics reason.
Reflections	(not described)	Page 4. Answer the three questions with concrete details from today.
Overall judgement	(not described)	Correct physics, the same symbols everywhere, tidy layout, nothing padded, inside the page limit.

3 Time plan, 13:00 to 17:00

Time	Block	Tasks
13:00 to 13:15	Start and calibrate	Connect the iOLab, press Reset twice, run the accelerometer, magnetometer and gyroscope calibration, do the sign check (Section 5.2), rezero the force probe, weigh the iOLab if there is a scale. One person starts the Part 1 sketch.
13:15 to 13:45	Part 1 runs	Two practice pushes, then 6 to 8 good runs. Export both CSV files after every run and move them into their run folder. Keep the run log.
13:45 to 13:55	Part 1 check	Run the notebook on Part 1. Redo any bad run now, while the setup still stands.
13:55 to 14:10	Part 2 setup	Build the ramp, measure H and L , sketch it.
14:10 to 14:40	Part 2 runs	6 to 8 good runs, export after every run.

Time	Block	Tasks
14:40 to 15:00	Analysis	Run the whole notebook. Choose the run for each figure. Copy the numbers into the tables.
15:00 to 16:20	Write pages 1 to 3	Split the work: one person on Part 1, one on Part 2, one on purpose, setup and sketch photos. Swap and read each other's parts at 16:00.
16:20 to 16:40	Page 4	Reflections.
16:40 to 16:55	Final check	Checklist in Section 9.8. Export to PDF.
16:55	Submit	Upload on DTU Learn. Do not wait until 17:00.

If you run late, do fewer runs and keep the analysis. Five good runs per part with uncertainties earn more than ten runs without them.

4 Physics and formulas

4.1 Signs and Newton's second law

Choose one axis, the y axis of the iOLab, and one positive direction. In Part 1 the positive direction is the direction of the push. In Part 2 it is up the ramp. Along that axis

$$\sum F_y = m a_y. \quad (1)$$

A force gets its sign from its direction, not from its name. Write your choice of positive direction in the report and draw it in the sketch.

4.2 What the accelerometer measures

The accelerometer does not measure $\vec{a}$. It measures $\vec{a} - \vec{g}$, which is the sum of all forces except gravity, divided by the mass. That is why it showed about 9.8 m/s^2 at rest on the table in Experiment 1, and about zero in free fall. Two cases matter today:

- iOLab lying flat (Part 1): gravity has no y component, so the reading is the real horizontal acceleration, $A_y = a_y$. The z reading is about $\pm 9.8 \text{ m/s}^2$; the sign depends on which face is down.
- iOLab on a ramp with y along the ramp, up positive (Part 2 extra): $A_y = a_y + g \sin \theta$.

4.3 Kinetic friction

When a surface slides on another, the friction force is $f_k = \mu_k n$ and points against the sliding. On a level table $n = mg$, so $f_k = \mu_k mg$. With no other horizontal force the acceleration is

$$a = -\mu_k g, \quad (2)$$

the same for any speed and any mass (to a good approximation). Before the sliding starts, static friction holds the iOLab until the push exceeds $\mu_s mg$.

4.4 Part 1 model: when does the hand let go?

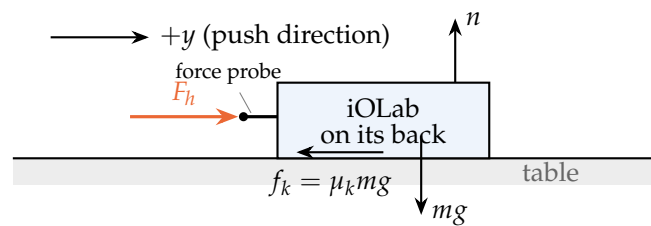


Figure 1. Forces on the iOLab while the hand pushes through the force probe. After the hand lets go, $F_h = 0$ and only f_k acts along y .

While the hand pushes and the iOLab slides:

$$ma = F_h - \mu_k mg \quad \implies \quad a = \frac{F_h}{m} - \mu_k g. \quad (3)$$

After the hand has let go, $F_h = 0$:

$$a = -\mu_k g = a_f \quad (\text{constant: the flat part of the graph}). \quad (4)$$

Solve equation (3) for the hand force and put in the three marked points:

$$F_h = m(a + \mu_k g) \quad (5)$$

Point	Acceleration	Hand force from (5)	Meaning
A	maximum, a_A	$F_A = m(a_A + \mu_k g)$, the largest	the hand pushes hardest
B	$a = 0$	$F_B = \mu_k mg \approx 0.4 \text{ N}$	the hand still pushes, exactly as hard as friction
C	$a = -\mu_k g$	$F_C = 0$	contact is lost

The number for B uses $m \approx 0.2 \text{ kg}$ and $\mu_k \approx 0.2$. So the theory says A and B are wrong and C is right. At B the speed is at its maximum ($a = 0$), which is why B feels like the moment the push ends. It is not: the hand still supplies the friction force.

Why does the acceleration fall from A to C? The hand slows down. As long as the hand slows down less than the free iOLab would ($\mu_k g$), it keeps pushing. When the hand brakes harder than that, the iOLab runs away from the finger. That moment is C. A hand that just touches can push but never pull, so $F_h \geq 0$ and the acceleration never goes below $-\mu_k g$ while the iOLab slides.

Figure 2 shows what to expect. It is made with a simple model ($m = 0.205 \text{ kg}$, $\mu_k = 0.18$, a smooth push). Your data will be noisier. Notice the start: the force rises before the acceleration does, because static friction holds the iOLab for the first 35 ms. The small bump near 6.0 s in the figure on the task sheet happens at the start of the push, long before the hand lets go, so it does not affect the question.

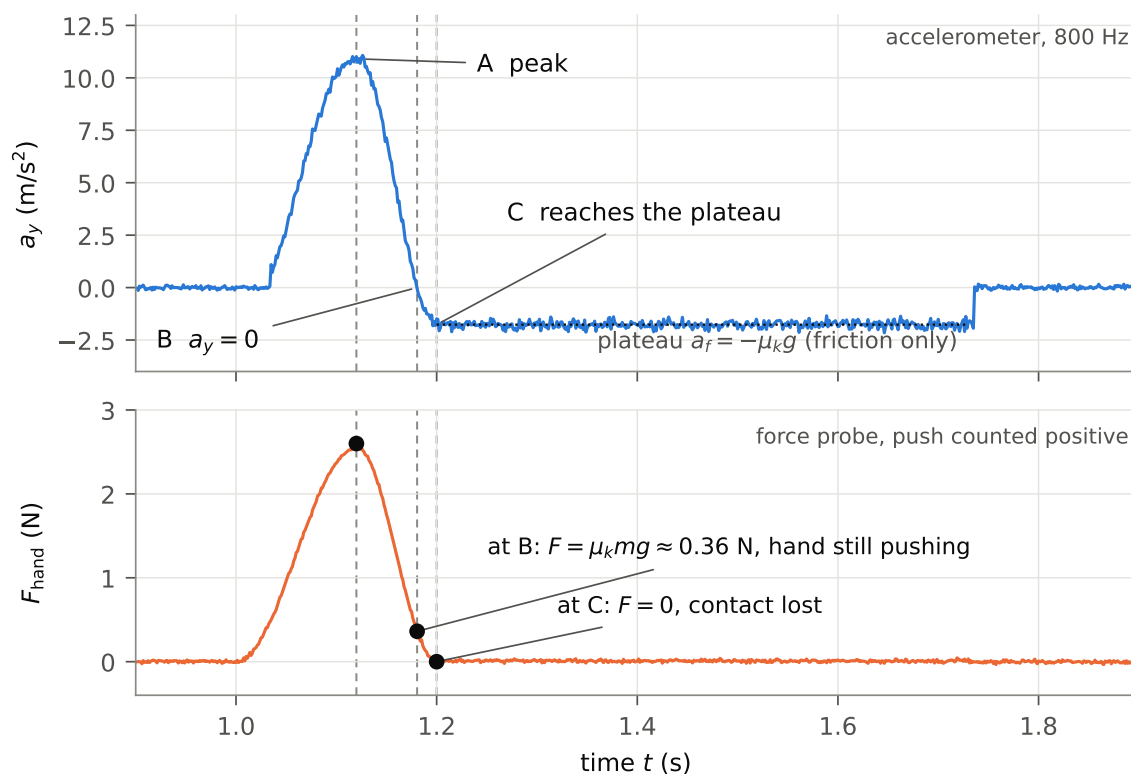


Figure 2. Simulated Part 1 run: accelerometer (top) and hand force (bottom) on the same time axis. At B the hand still pushes with $\mu_k mg \approx 0.36$ N. The force reaches zero at C, where the acceleration joins the friction plateau. Here A to B takes 60 ms and B to C only 20 ms.

4.5 Part 2 model: up and down the ramp

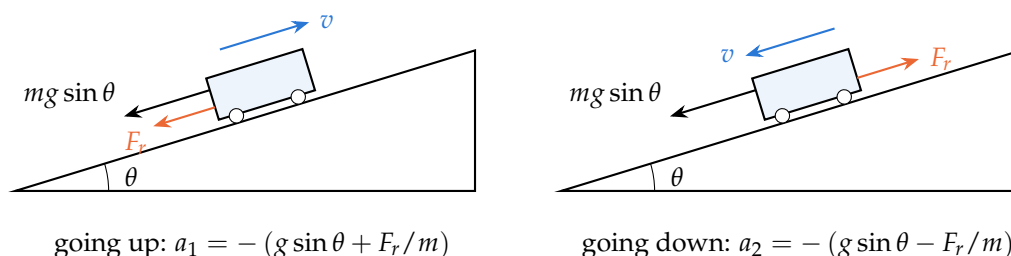


Figure 3. Forces along the ramp. Up the ramp is positive. Gravity pulls down the ramp both ways. The rolling resistance F_r opposes the velocity, so it flips direction at the turning point. The normal force and $mg \cos \theta$ act across the ramp and are left out.

With up the ramp positive, gravity along the ramp is $-mg \sin \theta$ both ways. The rolling resistance F_r (wheel bearings, the wheels rolling on the surface) points against the velocity: $-F_r$ going up, $+F_r$ going down.

$$\text{Up: } ma_1 = -mg \sin \theta - F_r \quad \Rightarrow \quad a_1 = -\left(g \sin \theta + \frac{F_r}{m}\right) \quad (6)$$

$$\text{Down: } ma_2 = -mg \sin \theta + F_r \quad \Rightarrow \quad a_2 = -\left(g \sin \theta - \frac{F_r}{m}\right) \quad (7)$$

Both are negative: the acceleration points down the ramp the whole time, also at the top where $v = 0$. For any $F_r > 0$ you get $|a_1| > |a_2|$. Adding and subtracting (6) and (7) gives two useful

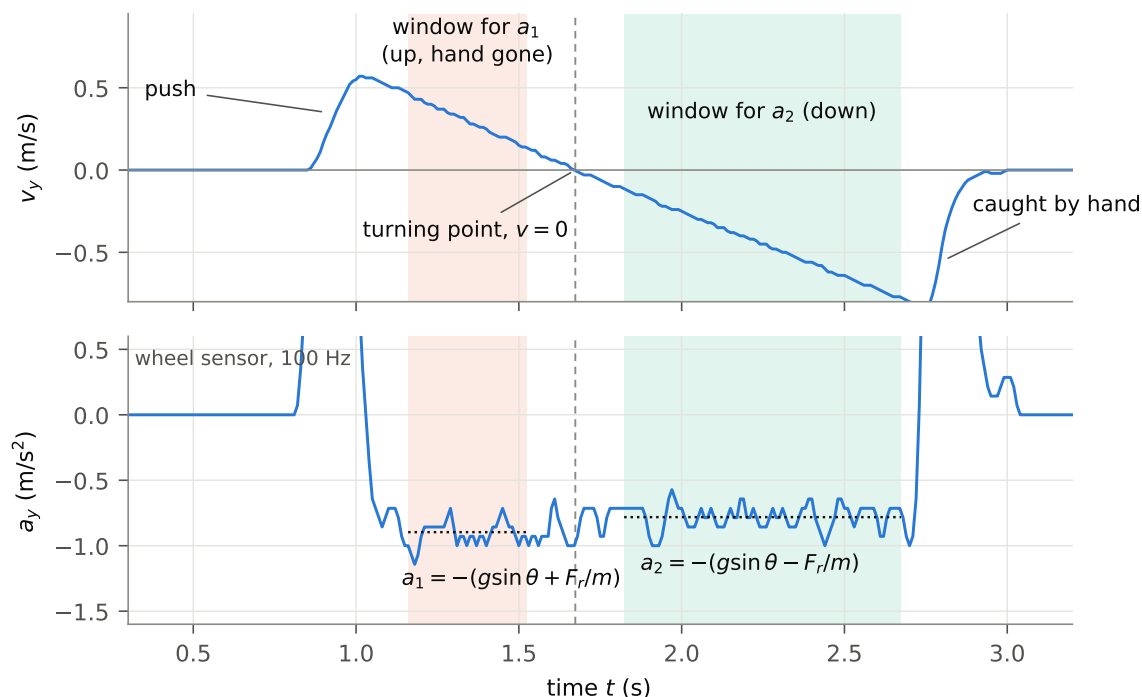


Figure 4. Simulated Part 2 run with the wheel sensor at 100 Hz ($\theta = 5^\circ$, $F_r/m \approx 0.06 \text{ m/s}^2$). The velocity is two straight lines with different slopes. The slope going up is a_1 , going down a_2 . The shaded windows skip the push, the turning point and the catch.

results:

$$\frac{|a_1| + |a_2|}{2} = g \sin \theta \quad \frac{|a_1| - |a_2|}{2} = \frac{F_r}{m} \quad (8)$$

The first has no friction in it, so you can test it against the angle you measure with a ruler, $\sin \theta = H/L$. The second gives the size of the rolling resistance. As a coefficient, $\mu_r = F_r/(mg \cos \theta)$.

Two small effects you can mention in the discussion:

- The wheels also have to spin up. That acts like a slightly larger mass, a few percent. Both $|a_1|$ and $|a_2|$ come out a bit smaller than (6) and (7), so $(|a_1| + |a_2|)/2$ lands a little below $g \sin \theta$. It does not change which of the two is larger.
- Air drag on the iOLab at 0.5 m/s is about 0.3 mN, which is about 0.002 m/s^2 . That is far below the rolling resistance, so you can leave it out.

4.6 Uncertainty toolkit

These are the rules from Experiment 1 and the course notebook on uncertainties. Use exactly these, and name them in the method section.

U1 One sample. A single reading has the uncertainty σ , the standard deviation of the signal on a quiet stretch.

U2 Mean of a selection in the app. Value μ , uncertainty $\sigma/\sqrt{N}$ with $N = f \Delta t$ (f sample rate, Δt length of the selection). Example: $\sigma = 0.10 \text{ m/s}^2$, $f = 100 \text{ Hz}$, $\Delta t = 0.40 \text{ s}$ gives $N = 40$ and $\sigma/\sqrt{N} = 0.016 \text{ m/s}^2$.

U3 Repeated runs. With n runs giving $x_1, \dots, x_n$:

$$\bar{x} = \frac{1}{n} \sum x_i, \quad s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}, \quad \delta \bar{x} = \frac{s}{\sqrt{n}} \text{ (SDOM)}.$$

Use the SDOM for the final a_1 and a_2 . It contains everything that changes from run to run. If it comes out smaller than a single run's uncertainty divided by $\sqrt{n}$, use that value instead (the notebook does this).

U4 *Sum or difference.* $\delta(x \pm y) = \sqrt{\delta x^2 + \delta y^2}$.

U5 *Product or quotient.* $\frac{\delta z}{|z|} = \sqrt{\left(\frac{\delta x}{x}\right)^2 + \left(\frac{\delta y}{y}\right)^2}$. Example: $\sin \theta = H/L$.

U6 *Function of one variable.* $\delta y = |f'(x)| \delta x$. Examples: $\mu_k = -a_f/g$ gives $\delta \mu_k = \delta a_f/g$, and $\theta = \arcsin(H/L)$ gives $\delta \theta = \delta(\sin \theta) / \cos \theta$ in radians.

U7 *Time.* A time read on a sampled graph is uncertain by about one sample, $\delta t = 1/f$. For a time difference between two sensors with rates f_a and f_F : $\delta t = \sqrt{1/f_a^2 + 1/f_F^2}$.

U8 *Value at an uncertain time.* If a graph has slope $\dot{y}$ at the moment you read it, $\delta y = \sqrt{\sigma^2 + (\dot{y} \delta t)^2}$. Read $\dot{y}$ as the slope s of a short selection in the app.

U9 *Standard score.* To compare two values:

$$z = \frac{|x_1 - x_2|}{\sqrt{\delta x_1^2 + \delta x_2^2}} \quad \begin{cases} z \leq 1 & \text{very likely the same} \\ 1 < z \leq 2 & \text{likely the same} \\ 2 < z \leq 3 & \text{grey zone} \\ z > 3 & \text{unlikely the same} \end{cases}$$

U10 *Rounding.* Round the uncertainty to one significant figure (two if it starts with 1), round the value to the same decimal place, write the unit. Example: $a_1 = -0.8953$ and $\delta a_1 = 0.0034$ become $a_1 = (-0.895 \pm 0.003) \text{ m/s}^2$.

5 The iOLab: settings that matter today

5.1 Start and calibration

1. Take the USB dongle out of the iOLab (press the metal end), plug it into the laptop, switch the iOLab on with the side button. The app shows the device at the top.
2. Press Reset twice. In the Tools menu, run the accelerometer, magnetometer and gyroscope calibration and follow the screen.
3. Force probe zero. Lay the iOLab the way you will use it in Part 1 (on its back), nothing touching the probe. Record 2 seconds, then click rezero under the force graph, as in Experiment 1. The notebook also uses the first second of every recording as the zero, so never touch the probe in that first second.
4. Force probe calibration with the 100 g mass (Tools menu) only if you have 10 minutes to spare. The main Part 1 test needs the zero of the force probe, not its scale. The scale matters only for the Newton check extra.
5. Weigh the iOLab if the lab has a scale. If not, use $m = (0.20 \pm 0.01) \text{ kg}$, a typical value for an iOLab with batteries, and write in the report that the mass is an estimate. Only the extras use m .

5.2 Sign check, one minute

- The coordinate axes are printed on both sides of the iOLab. Find $+y$.
- Tick Force, press Record, push the probe gently with a finger, press Stop. Note the sign. On many units a push reads negative and a pull positive, but check yours.
- Tick Wheel, roll the iOLab by hand toward $+y$. The velocity should be positive.

Write the result in the log. The notebook flips the signs so that the push (Part 1) and the motion up the ramp (Part 2) are positive. Use the same convention in the report and say so in one sentence.

5.3 Sample rate and smoothing

- The app picks the sample rates from the sensors you tick. Each chart title shows its rate, for example “Accelerometer (800 Hz)”. The accelerometer can run up to 800 Hz, the force probe faster, the wheel at 100 Hz.
- Tick only the sensors you need. Fewer sensors usually give higher rates.
- Write the rates down. They go into the report and into $\delta t = 1/f$.
- Set the data smoothing to its lowest setting (no smoothing) for Part 1. Smoothing rounds the sharp corner at C and moves it in time. It also makes σ look smaller than the real noise.

5.4 Reading numbers in the app

- With the graph icon active, click and drag over a region. The box shows μ (mean), σ (standard deviation), a (area under the curve) and s (average slope, with R^2 in brackets). The length of the selected interval is shown too; that is your Δt .
- A selection that covers one sample gives the value of that sample.
- Zoom with the magnifier by dragging a rectangle, double click to zoom out. All charts share the time axis, so a zoom on the accelerometer also zooms the force graph. This is how you read two sensors at the same moment.

5.5 Saving and exporting

- Every recording is saved by itself in Documents/iOLab-WorkFiles/rawdata. The folder icon opens old runs.
- To export a chart: click the toolbar icon at the bottom left of that chart, then Export to CSV. The file goes to Documents/iOLab-WorkFiles/export, named after the time and the sensor. The columns are time and `cal[0]`, `cal[1]`, `cal[2]` (x , y , z for the accelerometer; the force probe only has `cal[0]`).
- Right after each run, move the new files into their own folder: data/part1/run1, then data/part1/run2, and so on. It takes 20 seconds and keeps runs from getting mixed up later.

6 Part 1 step by step

6.1 Purpose

Find out when the pushing hand loses contact with the sliding iOLab. You test one hypothesis with a direct measurement of the contact force. Say this in your own words in one or two sentences.

6.2 Which hypothesis, and how to make it testable

Choose hypothesis C: the hand loses contact when the acceleration flattens out. Theory predicts it (Section 4.4), and the same data also show that A and B fail, which gives the strongest conclusion.

A testable hypothesis names four things. For C they are:

1. The claim: contact ends at point C.
2. What you measure: the moment the hand force returns to zero, t_{F0} , with the force probe, and the acceleration at that moment, $a(t_{F0})$, with the accelerometer.
3. The prediction: $a(t_{F0})$ equals the plateau value a_f . At t_A and t_B the force is still clearly above zero.
4. The decision rule: C is not rejected if the standard score between $a(t_{F0})$ and a_f is $z \leq 2$. It is rejected if $z > 3$.

Your group may prefer to test B, which is the common intuition. The experiment is the same. The outcome is then “rejected”.

6.3 Setup

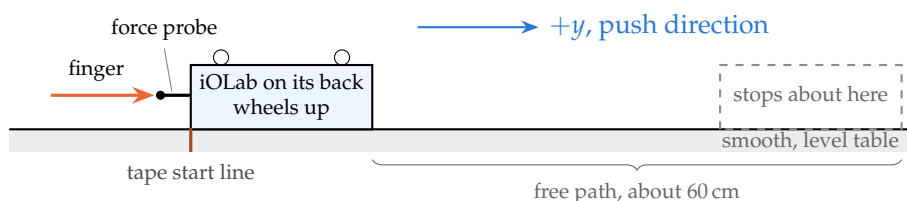


Figure 5. Part 1 setup seen from the side. Your hand drawn sketch needs the same labels, plus the laptop and the sensors you ticked with their sample rates.

- A smooth, clean, level table with a free path of about 60 cm. Avoid seams and edges.
- The iOLab on its back, y axis along the push, force probe toward you. Mark a start line with tape.
- Push only through the force probe, with one fingertip on the probe tip. If your finger slips, push with the end of a pencil held in the hand. A push on the body of the iOLab is not measured by the probe and ruins the test.
- Sensors: tick Accelerometer and Force, nothing else. Smoothing off. Note both sample rates.

6.4 One run, in this order

1. iOLab on the start line, finger off the probe. Press Record.
2. Wait one second. This quiet stretch sets the zero of both sensors.
3. Put your fingertip on the probe and push smoothly along y for about 0.2 s. Then let your finger slow down gently, as if you follow the iOLab, until it runs away from the finger. Do not jab.
4. Let it slide and stop by itself. Wait one second. Press Stop.
5. Look at the run. You want one clean peak in a_y , a flat friction level, a force that goes back to zero and stays there. If the iOLab turned, hit something or the finger slipped off the probe, delete the run and do it again.
6. Export the accelerometer chart and the force chart. Move both files into `data/part1/runN`. Write the run in the log.

Get 6 to 8 good runs. Vary the push a little between runs (softer, harder). That shows the result does not depend on how you push.

Why the gentle release: B and C are close in time. In the simulated run they are only 20 ms apart, two samples at 100 Hz. A smooth release stretches the time between B and C, so it is easy to resolve even at a low sample rate.

Plan B, if the app will not run Force and Accelerometer together: film the push from the side in slow motion (240 frames per second) with a phone held close to the probe. Before the push, tap the iOLab once lightly; the tap is a spike in a_y and a visible touch in the video, so it lines up the two clocks. The frame where the finger leaves the probe gives t_{F0} to about 4 ms.

6.5 What to read and how

The notebook finds all of these by itself. If you work in the app instead (or want to check the notebook), read them like this:

Quantity	Graph	How
Force noise σ_F	force	Select the quiet first second. $\mu \approx 0$, read σ_F .
t_A, a_A (point A)	acceleration	Zoom on the peak, select the highest sample.
t_B (point B)	acceleration	The first sample after the peak where a_y crosses zero.
$a_f, \delta a_f$ (plateau)	acceleration	Select the middle of the flat part, 0.2 to 0.5 s long. $a_f = \mu$, $\delta a_f = \sigma / \sqrt{f\Delta t}$ (rule U2).
t_C (point C)	acceleration	The first sample after B inside the band $a_f \pm 2\sigma$.
t_{F0} (contact lost)	force	The first sample after the peak where the force is back inside $\pm 2\sigma_F$.
$a(t_{F0})$	acceleration	Select the samples within about 5 ms of t_{F0} . Value μ , uncertainty $\sigma / \sqrt{n}$, plus the timing term of rule U8 if the graph is steep there.
$F(t_B)$ (extra)	force	The force at t_B , uncertainty by rule U8.

6.6 The decision

For every run, and for the mean over all runs, compute three standard scores (rule U9):

$$z_C = \frac{|a(t_{F0}) - a_f|}{\sqrt{\delta a(t_{F0})^2 + \delta a_f^2}}, \quad z_B = \frac{|a(t_{F0}) - 0|}{\delta a(t_{F0})}, \quad z_A = \frac{|a(t_{F0}) - a_A|}{\sqrt{\delta a(t_{F0})^2 + \sigma^2}}. \quad (9)$$

Hypothesis C is not rejected if $z_C \leq 2$. Hypotheses A and B are rejected if $z_A > 3$ and $z_B > 3$. Back it up with the times: $t_{F0} - t_B$ should be tens of milliseconds and clearly positive, while $t_{F0} - t_C$ should be within one or two samples of zero. The C criterion ($a_f \pm 2\sigma$) fires a little early on purpose, so a small positive $t_{F0} - t_C$ is normal.

6.7 Extras that lift the report

- Friction coefficient from the plateau: $\mu_k = -a_f/g$, $\delta\mu_k = \delta a_f/g$.
- Force at B against theory: equation (5) gives $F_B = \mu_k mg$. A measured $F(t_B)$ that matches is strong support for the whole model.
- Newton check: while the hand pushes, $a = F_h/m - \mu_k g$ (equation (3)). A straight line fit of a against F has slope $1/m$ and intercept $-\mu_k g$. Compare the fitted mass with the scale. A few percent off is normal if the force probe was not calibrated today.

6.8 Rough values to catch mistakes

These are ranges to spot errors, not targets.

Quantity	Rough value
Peak a_A	5 to 15 m/s ²
Plateau a_f	-1 to -3 m/s ² (μ_k from 0.1 to 0.3)
Sliding time after the push	0.2 to 1 s
$F(t_A)$	1 to 4 N
$F(t_B) = \mu_k mg$	0.2 to 0.6 N
$t_{F0} - t_B$	10 to 100 ms
$a(t_{F0})$	equal to a_f within its uncertainty

6.9 Mistakes that cost points

- Pushing on the body instead of the probe. The force stays at zero and the test says nothing.

- Smoothing left on. The corner at C moves and σ shrinks.
- Touching the probe in the first second. The zero is wrong, and the force never seems to reach zero.
- The iOLab rotates or drifts sideways during the slide. Then part of the motion is along x . Redo the run.
- Reporting one run only. Show all runs in the table and give the mean with its uncertainty.
- Calling A, B or C “correct” without a number. Every verdict needs a z value.

7 Part 2 step by step

7.1 Purpose

Compare the acceleration going up the ramp, a_1 , with the acceleration coming down, a_2 , including their uncertainties, and explain the difference with the forces that act.

7.2 The hypothesis

Recommended: $|a_1| > |a_2|$. The iOLab slows down faster going up than it speeds up going down. The four parts:

1. The claim: $|a_1| > |a_2|$, because rolling resistance adds to gravity going up and works against it going down (Section 4.5).
2. What you measure: a_1 and a_2 with the wheel sensor, in 6 to 8 runs, each averaged over the runs with its SDOM.
3. The prediction: $D = |\bar{a}_1| - |\bar{a}_2|$ is positive.
4. The decision rule: with $\delta D = \sqrt{\delta \bar{a}_1^2 + \delta \bar{a}_2^2}$ and $z = D/\delta D$, the hypothesis is not rejected if $z > 3$ (with $D > 0$). It is rejected if $z \leq 2$, because then a_1 and a_2 are the same within the uncertainty. Between 2 and 3 you cannot decide; say so.

An alternative is the hypothesis you would make without friction: “ $a_1 = a_2$ within the uncertainty, $z \leq 2$ ”. You will most likely reject it. Both versions work for the rubric as long as the decision rule is stated. Pick one and stick to it.

Keep the signs straight in the report: with up the ramp positive, both a_1 and a_2 are negative. The hypothesis compares their sizes.

7.3 Setup

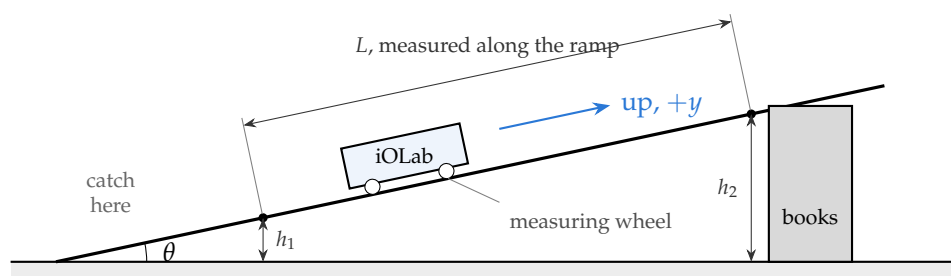


Figure 6. Part 2 setup seen from the side, angle drawn larger than real. $H = h_2 - h_1$ and $\sin \theta = H/L$. The measuring wheel is the single wheel.

- A straight ramp of at least 1 m, raised at one end to about 3° to 5° . That is a rise of 5 to 9 cm per metre. A small angle gives a long, slow trip (more data) and makes friction a larger part of the effect. If the iOLab does not roll back down by itself, make it steeper.
- Mark two points on the ramp with tape, far apart. Measure L between them along the ramp and the heights h_1 and h_2 of the ramp surface above the table at the marks. $H = h_2 - h_1$, $\sin \theta = H/L$. Give each length an uncertainty (reading plus placing, about 1 to 2 mm).

- Optional second angle: lay the iOLab still on the ramp with y along the slope and record the accelerometer for 3 seconds. Then $\theta = \arctan(|A_y|/|A_z|)$ with the means of the two readings.
- iOLab on its wheels, y axis along the ramp, measuring wheel on the ramp surface, straight path. Use a guide edge if the ramp has one.
- A catch zone at the bottom: your hand, or a soft stop (a sponge or a folded cloth against a book).
- Sensors: tick Wheel (position, velocity and acceleration). The accelerometer is an optional extra. Note the rates (the wheel runs at 100 Hz).

7.4 One run, in this order

1. iOLab at the start near the bottom, held still. Press Record.
2. Hold it still for one second. (The notebook uses this still second for the angle if you record the accelerometer.)
3. Give a short push up the ramp and take your hand away at once. It should travel 30 to 60 cm up. Longer trips give better data.
4. Let it roll up, stop and roll back. Catch it gently before it hits anything. Press Stop.
5. Look at the run. The velocity should be two straight lines that cross zero once. If it hit the top, slid sideways or got pushed during the trip, redo it.
6. Export the wheel chart (and the accelerometer chart if ticked) into data/part2/runN. Write the run in the log.

Get 6 to 8 good runs, starting from the same mark each time.

7.5 What to read and how

The notebook does this for you. By hand in the app:

- Up window, on the acceleration graph: start where the curve has settled after the push (the Part 1 lesson: the hand is gone only when the curve is flat) and stop about 0.15 s before the velocity reaches zero. Read μ , σ and Δt . Then $N = f\Delta t$ and $\delta a_1 = \sigma/\sqrt{N}$ for that run.
- Down window: from about 0.15 s after the velocity crosses zero to about 0.1 s before the catch. That gives a_2 .
- Cross check: select the same windows on the velocity graph. The slope s should match μ from the acceleration graph.
- Stay away from the turning point. Near $v = 0$ the wheel turns less than one millimetre per sample and the friction flips direction, so the acceleration graph jumps there.

7.6 The decision

1. Average over the runs: $\bar{a}_1 \pm \delta\bar{a}_1$ and $\bar{a}_2 \pm \delta\bar{a}_2$ (rule U3).
2. $D = |\bar{a}_1| - |\bar{a}_2|$, $\delta D = \sqrt{\delta\bar{a}_1^2 + \delta\bar{a}_2^2}$ (rule U4), $z = D/\delta D$.
3. Apply your decision rule and write the verdict with the numbers.

7.7 Theory check and extras

- Friction cancels in the mean: compare $(|\bar{a}_1| + |\bar{a}_2|)/2$ with $g \sin \theta$ from the ruler (equation (8)), with a standard score. Expect the measured value to sit slightly below $g \sin \theta$ because the wheels spin (Section 4.5).
- Size of the rolling resistance: $F_r/m = (|\bar{a}_1| - |\bar{a}_2|)/2$, then F_r and $\mu_r = F_r/(mg \cos \theta)$.
- Paired check: compute $|a_1| - |a_2|$ inside each run first, then its mean and SDOM over the runs. Anything that changes from run to run but hits a_1 and a_2 alike (a harder push, a slightly different start point) cancels. The notebook prints both versions.
- Extra with the 3D accelerometer: along the ramp it reads $A_y = a_y + g \sin \theta$ (Section 4.2). Insert a_1 and a_2 : going up $A_y = -F_r/m$, going down $A_y = +F_r/m$. The accelerometer shows the friction directly, and the jump of A_y at the turning point is the friction changing direction.

The difference of the two readings should equal $|a_1| - |a_2|$.

7.8 Rough values to catch mistakes

Quantity	Rough value (for $\theta = 4^\circ$, so $g \sin \theta = 0.685 \text{ m/s}^2$)
$ a_1 $	a little above $g \sin \theta$, for example 0.72 to 0.85 m/s^2
$ a_2 $	a little below $g \sin \theta$, for example 0.55 to 0.66 m/s^2
$ a_1 - a_2 = 2F_r/m$	0.05 to 0.3 m/s^2
Time going up	0.6 to 1.2 s
SD of a_1 between runs	0.005 to 0.03 m/s^2

7.9 Mistakes that cost points

- Starting the a_1 window while the hand still pushes. If the curve has not settled, the push is still going, and $|a_1|$ comes out too small.
- Windows across the turning point. They mix a_1 and a_2 .
- Comparing a_1 and a_2 with signs mixed up. State the positive direction once, then compare magnitudes.
- Using σ instead of $\sigma/\sqrt{N}$ for a mean, or the other way round.
- A final result without the uncertainty of the mean over runs.
- Angle measured once with no uncertainty.

8 The analysis notebook

Experiment3_analysis.ipynb does every calculation in this guide on your exported files. It was tested on simulated iOLab files in three export layouts. It prints numbers, tables and figures. It writes no report text.

Before the lab (5 minutes, do it now)

1. Put the notebook in a folder, for example your 3rd_experiment folder.
2. Make the folders data/part1 and data/part2 next to it.
3. Open it in Jupyter or VS Code and run all cells once. With no data it should print “No Part 1 runs analysed” and stop without errors. If a package is missing, run this in a terminal:

```
pip install numpy pandas matplotlib
```

In the lab

1. One folder per run: data/part1/run1, run2, and so on, each with the accelerometer CSV and the force CSV of that run. For Part 2: data/part2/run1, and so on, each with the wheel CSV (and the accelerometer CSV if you recorded it). The file names inside a run folder do not matter.
2. In the settings cell, fill in the mass and uncertainty, and H, dH, L, dL in metres.
3. Run all cells. Look at the plot of every run. Check that A, B and C sit where you would put them by eye, and that the Part 2 windows avoid the push, the turning point and the catch.
4. If a Part 2 window is wrong, set it by hand in the settings cell and run again, for example

```
MANUAL_WINDOWS = {"run3": {"up": (1.20, 1.55), "down": (1.80, 2.60)}}
```

What it gives you

- Part 1: a table with $t_A, t_B, t_C, t_{F0}, a_f, a(t_{F0})$, the three z values, $F(t_B), F(t_C)$ and μ_k for every run, then the summary over runs and the Newton check.
- Part 2: a table with a_1 and a_2 per run by two methods (slope of the velocity, and mean of the acceleration graph as in the app), the summary with $D, \delta D$ and z , the angle check and F_r .

- Report figures in the output folder at 300 dpi (PNG and PDF): `part1_report_figure`, `part2_report_figure`, `part2_runs_figure`, plus one diagnostic plot per run.

If the CSV export fails

Read the numbers in the app (Sections 6.5 and 7.5) and type them into `MANUAL_PART1` and `MANUAL_PART2` in the settings cell. The settings cell shows the format. The notebook then computes the same summaries.

If the notebook reads a file wrongly

It prints how it read every file. If the push was not along y , set `ACC_AXIS = "y"`. If a wheel file with a single column is read as the wrong quantity, set `WHEEL_SINGLE_KIND` to `"position"`, `"velocity"` or `"acceleration"`.

9 Report blueprint

The rubric rewards clear evidence, not length, and three pages fill up fast. This section says what goes where and what each part must contain. The words are yours.

9.1 Page plan

Page	Content
1	Title block. Purpose of both parts (3 to 4 lines). Part 1 setup: sketch at about a third of the page width, next to 3 lines of setup text (orientation, sensors and rates, surface, mass). Hypothesis C with its decision rule. Method in 4 to 6 lines.
2	Part 1 figure (one run, a_y and F against time, A, B, C marked) and the table of runs. Results in 3 to 4 lines with numbers. Discussion and verdict in 5 to 7 lines. If space is left: Part 2 sketch and hypothesis.
3	Part 2 setup and method, the table of runs with means, one figure, results with z , the angle check, the discussion and verdict, and 3 to 4 lines on error sources.
4	Reflections only.

Space savers: put the sketch and a graph side by side; use a smaller font inside tables; do not copy the task text; one idea per sentence.

9.2 What each section must contain

Title block

- ☐ A title that says what you tested, not only "Experiment 3".
- ☐ Names, study numbers, group number, date, course number.

Purpose

- ☐ One sentence per part: what you tested and with which sensors.
- ☐ No background story. The conveyor belt text on the task sheet is not needed.

Setup, for each part

- ☐ The hand drawn sketch with labels (Section 9.3).
- ☐ Orientation of the iOLab, the positive direction, the sensors you ticked and their sample rates.
- ☐ Part 1: the surface, that the push goes through the force probe, the mass with its uncertainty.
- ☐ Part 2: L , H and θ with uncertainties, the ramp, where you caught the iOLab.

Hypothesis, for each part

- ☐ The four elements: claim, measured quantities, prediction, decision rule with z .
- ☐ Placed before the results. It is a prediction, so it must not read as if you wrote it after seeing the data.

Method, for each part

- ☐ One run in 2 or 3 sentences: start, push, stop.
- ☐ Number of runs, and why you repeated.
- ☐ How each quantity came out of the data: the definitions of A , B , C and t_{F0} , or the two windows for a_1 and a_2 .
- ☐ The uncertainty methods by name: $\sigma/\sqrt{N}$ with $N = f\Delta t$, SDOM over runs, propagation, standard score with its limits.
- ☐ If you used the notebook, one sentence that the analysis was done in Python on the exported CSV files.

Results, for each part

- ☐ One table with every run, and the mean with its uncertainty (templates below).
- ☐ One figure with axis labels, units, marked points or windows, and a caption that says what it shows, which run, and what to look at.
- ☐ The deciding numbers again in the text, with uncertainty and unit.

Discussion and conclusion, for each part

- ☐ The verdict word: “not rejected” or “rejected”. Avoid “proven”.
- ☐ The numbers that decide: the two values you compare, their uncertainties, z , and the limit you set.
- ☐ The physics reason, with the equations. Part 1: equations (3) and (5), so the hand force is zero exactly when the acceleration equals $-\mu_k g$. Part 2: equations (6) and (7), friction changes direction at the turning point.
- ☐ Part 1: why A and B fail. The force is still clearly above zero there, and $F_B = \mu_k mg$.
- ☐ Part 2: the angle check and the size of F_r .
- ☐ One or two error sources, the direction in which each pushes the result, and whether it could change the verdict (list below).

9.3 The sketch

Draw it by hand on paper. Figures 5 and 6 show the level of labelling.

- Part 1:
 - ☐ table surface and the free path
 - ☐ iOLab on its back, wheels up
 - ☐ force probe, and the finger pushing it
 - ☐ the $+y$ axis and the push direction
 - ☐ laptop with dongle, drawn small, with the sensors and sample rates written next to it
- Part 2:
 - ☐ ramp, its support and the angle θ
 - ☐ the two marks, L along the ramp, h_1 and h_2
 - ☐ iOLab on its wheels, with the measuring wheel marked
 - ☐ $+y$ up the ramp, the path up and back, and arrows for a_1 and a_2 , both pointing down the ramp
 - ☐ the catch zone

Photo: daylight, camera straight above the paper, crop the edges, darken the lines if needed. The text in the sketch must be readable at the size you insert it.

9.4 Table templates

Part 1, one row per run, the mean in the last row. Units in the header, not in the cells.

Run	f_a (Hz)	t_A (s)	t_B (s)	t_C (s)	t_{F0} (s)	a_f (m/s ²)	$a(t_{F0})$ (m/s ²)	z_C	z_B
1									
2									
⋮									
Mean									

Part 2, one row per run, then the mean and its SDOM, then the comparison.

Run	a_1 (m/s ²)	δa_1 (m/s ²)	a_2 (m/s ²)	δa_2 (m/s ²)	$ a_1 - a_2 $ (m/s ²)
1					
2					
⋮					
Mean $\pm$ SDOM					

Below it, in one line: $D = |\bar{a}_1| - |\bar{a}_2|, \delta D, z$. And a small table for the angle: $h_1, h_2, H, L, \sin \theta, \theta, g \sin \theta$, all with uncertainties.

9.5 Figures

- ☐ Axis labels with symbol and unit, for example “ a_y (m/s²)” and “time t (s)”.
- ☐ Points A, B, C marked; the moment the force reaches zero marked; windows shaded.
- ☐ A caption of about two lines: what is shown, which run, what the reader should see.
- ☐ The text refers to every figure (“Figure 2 shows ...”).
- ☐ Readable when printed. No text smaller than the caption.

9.6 Error sources worth one sentence each

Source	Effect and direction
<i>Part 1</i>	
Sample rate	A time is only known to about $1/f$. It limits how well t_{F0} and t_C can be told apart, for both sensors alike.
Force probe zero	An offset makes the force look nonzero after contact. The quiet first second corrects it.
Friction not constant	Dust or speed dependence tilts the plateau slightly. It widens σ but does not move C.
Rotation of the iOLab	Part of the motion goes along x . Such runs were discarded.
<i>Part 2</i>	
Hand still touching at the start of the up window	Makes $ a_1 $ too small, so it hides the difference. The verdict is then on the safe side.
Angle	Matters for the $g \sin \theta$ check only, not for comparing a_1 and a_2 .
Wheel resolution near $v = 0$	The graph jumps at the turning point. Excluded by the windows.
Spinning wheels	Both $ a $ a few percent smaller. The order stays the same.

Source	Effect and direction
Ramp not straight	a changes along the ramp. Same start mark in every run keeps it the same for all runs.

9.7 Page 4: reflections

Three to four lines per question. Be concrete and mention a moment from today. These are prompts to think about, not answers to copy.

1. *How did you approach the experiment: did you expect a result, or did it come from the data?* Did the group expect B, the moment of maximum speed? Did anyone disagree before the first run? For Part 2, did you predict the sign of $|a_1| - |a_2|$ before you saw a graph?
2. *What did you learn?* Pick one or two concrete things. For example: that the push lasts beyond the moment of maximum speed; that a second sensor (the force probe) can test what one sensor (the accelerometer) cannot; that friction changes direction at the top of the ramp; that the standard score, not the look of a graph, decides.
3. *What do you think you were supposed to learn?* Your view of the aim: writing a hypothesis that can fail, designing a measurement that can reject it, and letting the uncertainty decide.

9.8 Final check before you submit

- ☐ At most 3 pages, then page 4.
- ☐ Names, study numbers, group, date, course.
- ☐ Purpose of both parts.
- ☐ Two hand drawn sketches, labelled and readable.
- ☐ Two hypotheses, each with a decision rule.
- ☐ Every number with uncertainty and unit, rounded by rule U10.
- ☐ Tables with every run; means with their SDOM.
- ☐ Figures with axis labels, units and captions, each referred to in the text.
- ☐ Two verdicts, each with a z value and a physics reason.
- ☐ Sample rates and numbers of runs stated.
- ☐ Reflections: three questions, 3 to 4 lines each.
- ☐ The same symbols everywhere ($a_1, a_2, a_f, \mu_k, \theta$).
- ☐ Exported to PDF; figures sharp.

A Formula sheet

Part 1, iOLab sliding on its back, push direction positive

While the hand pushes	$a = \frac{F_h}{m} - \mu_k g$ and $F_h = m(a + \mu_k g)$
After contact is lost	$a_f = -\mu_k g$ (constant), $\mu_k = -a_f/g$, $\delta\mu_k = \delta a_f/g$
Hand force at A, B, C	$F_A = m(a_A + \mu_k g)$, $F_B = \mu_k mg$, $F_C = 0$
Test of C	$z_C = \frac{ a(t_{F0}) - a_f }{\sqrt{\delta a(t_{F0})^2 + \delta a_f^2}} \leq 2$ (A and B: $z > 3$)
Newton check	fit $a = kF + c$: $m = 1/k$, $\delta m = \delta k/k^2$, $c = -\mu_k g$

Part 2, iOLab on wheels on a ramp, up the ramp positive

Going up, going down	$a_1 = -\left(g \sin \theta + \frac{F_r}{m}\right)$, $a_2 = -\left(g \sin \theta - \frac{F_r}{m}\right)$
Sum and difference	$\frac{ a_1 + a_2 }{2} = g \sin \theta$, $\frac{ a_1 - a_2 }{2} = \frac{F_r}{m}$, $\mu_r = \frac{F_r}{mg \cos \theta}$
Angle from a ruler	$\sin \theta = \frac{H}{L}$, $\frac{\delta(\sin \theta)}{\sin \theta} = \sqrt{\left(\frac{\delta H}{H}\right)^2 + \left(\frac{\delta L}{L}\right)^2}$, $\delta \theta = \frac{\delta(\sin \theta)}{\cos \theta}$
Angle from the accelerometer at rest	$\theta = \arctan \frac{ A_y }{ A_z }$
Accelerometer along the ramp (extra)	$A_y = a_y + g \sin \theta$: going up $A_y \approx -F_r/m$, going down $A_y \approx +F_r/m$
Test	$D = \bar{a}_1 - \bar{a}_2 $, $\delta D = \sqrt{\delta \bar{a}_1^2 + \delta \bar{a}_2^2}$, $z = D/\delta D$

Uncertainty

Mean of a selection	$\mu \pm \sigma/\sqrt{N}$, $N = f \Delta t$
Repeated runs	$\bar{x} = \frac{1}{n} \sum x_i$, $s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$, $\delta \bar{x} = s/\sqrt{n}$
Sum, difference	$\delta = \sqrt{\delta x^2 + \delta y^2}$
Product, quotient	$\delta z/ z = \sqrt{(\delta x/x)^2 + (\delta y/y)^2}$
Function	$\delta y = f'(x) \delta x$
Time	$\delta t = 1/f$; two sensors: $\delta t = \sqrt{1/f_a^2 + 1/f_F^2}$
Value at an uncertain time	$\delta y = \sqrt{\sigma^2 + (\dot{y} \delta t)^2}$
Standard score	$z = x_1 - x_2 / \sqrt{\delta x_1^2 + \delta x_2^2}$: ≤ 1 very likely same, ≤ 2 likely same, ≤ 3 grey zone, > 3 unlikely same
Constants	$g = 9.82 \text{ m/s}^2$ (the course value), $m_{\text{iOLab}} \approx 0.20 \text{ kg}$ (weigh it)

B Run log

Copy this onto paper, one line per recording, bad runs included. It saves you in the analysis.

Part	Run	File time stamp	Good?	Notes (push, what went wrong, rates)
1	1			
1	2			
1	3			
1	4			
1	5			
1	6			
1	7			
1	8			
2	1			
2	2			
2	3			
2	4			
2	5			
2	6			
2	7			
2	8			

Setup numbers: m = _____ h_1 = _____ h_2 = _____ L = _____

Sample rates: accelerometer _____ Hz force _____ Hz wheel _____ Hz

Sign check: push on the probe reads _____ rolling toward $+y$ reads _____